

$$\begin{cases} 1 - \theta, x = 0 \\ \frac{\theta}{2}, x = \pm 1 \end{cases}$$

write as exponential family, show family is full-ranked, shows that sufficient statistic is also complete...

$\mathbb{P}[\phi(x) = 0] < 1$ expectation is zero, but random variable itself is not always zero

Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{N}(\theta, \theta)$, and $\theta > 0$ is an unknown parameter. Determine a complete sufficient statistic for θ and carefully justify your answer. start by checking whether this is an exponential family or not...

Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{N}(\theta, \theta^2)$, and $\theta > 0$ is an unknown parameter.

1. For $n = 1$, show that $T = X$ is a minimum sufficient statistic but is not a complete sufficient statistic.
2. Show that the sufficient statistic

$$T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is not complete for any $n \geq 2$.

able to compute $\theta[(2x_i)^2] \theta[\sum X_i^2]$